CSA15: Cloud Computing and Big Data Analytics

Scenario-Based Assessment Tool 2: Answers for Set 1, Set 2 and Set 3

SET 1 (CO2: Virtualization, Performance and Security)

Q1. Video streaming platform for millions of concurrent users

Architecture

- **Ingest and transcoding:** Uploaded masters go to object storage (S3 or Blob). A queue-driven fleet of VM or container workers transcodes each video into adaptive-bitrate renditions (HLS/DASH at 240p to 4K).
- **Delivery:** A global CDN with edge caching serves video segments, so most requests never reach the origin. An origin-shield tier protects storage from cache-miss storms.
- **Application tier:** Stateless microservices (catalog, auth, recommendations, playback) run in containers behind load balancers. Redis/Memcached caches metadata and sessions.
- **Data tier:** A managed NoSQL database for user and watch data, with read replicas and sharding for scale.

Load balancing and auto-scaling

- A global DNS or anycast load balancer routes users to the nearest healthy region. Layer-7 load balancers spread traffic across instances with health checks, so failed nodes are removed automatically.
- Auto-scaling groups add or remove instances based on CPU, requests per second and concurrent streams. Scheduled scaling pre-warms capacity for known peaks such as premieres and live events. Because services are stateless, new instances serve traffic immediately, which keeps buffering low.

Cost optimization

- Reserved or committed-use pricing for the steady baseline, spot instances for fault-tolerant transcoding, and on-demand only for bursts.
- Storage tiering (hot to infrequent to archive) for older titles, and high CDN cache-hit ratios to cut egress cost.
- Right-sizing, per-title encoding (fewer bits for the same quality), and budgets and alerts to catch waste early.

Q2. Containers across multiple cloud providers (fintech)

Solution

- Package every service as a container image stored in a registry that all clouds can pull from. Run **Kubernetes** (EKS, AKS, GKE, or a management layer such as Anthos or Rancher) in each cloud so the deployment model is the same everywhere.

- Use **GitOps** (Argo CD or Flux) with Helm charts, so Git is the single source of truth and every cluster converges to the declared state. Infrastructure as code (Terraform) keeps cloud resources consistent.
- Add policy-as-code (OPA/Gatekeeper), image scanning and signed images to keep security uniform.

Service discovery and scaling

- Kubernetes DNS and Services give every workload a stable name. A service mesh (Istio or Linkerd) adds mTLS, retries and cross-cluster routing.
- The Horizontal Pod Autoscaler scales pods on CPU or custom metrics, and the Cluster Autoscaler adds nodes when pods cannot be scheduled.

Benefits and risks

- **Benefits:** avoids vendor lock-in, improves resilience, lets the company pick the best service or price per cloud, and helps with data-residency rules.
- **Risks:** higher operational complexity, skills gaps, inter-cloud egress cost, inconsistent IAM and security models, harder monitoring and compliance, and latency between clouds.

Q3. Backup and disaster recovery (e-learning platform)

Design

- **Versioning:** Enable object versioning and soft-delete on storage, plus MFA-delete or object lock, so an accidental deletion only adds a delete marker and the earlier version can be restored.
- **Backups:** Automated, policy-based snapshots of databases and volumes (for example AWS Backup) with retention rules, stored in a separate account with restricted delete rights.
- **Replication:** Cross-region replication of objects and database read replicas, so a regional failure does not remove the only copy.
- **Least privilege:** IAM roles and approval workflows limit who can delete production data.

Achieving RTO and RPO

- **RPO** (acceptable data loss) is met by backup and replication frequency. Continuous or near-real-time replication gives an RPO of seconds to minutes, while daily snapshots give 24 hours.
- **RTO** (acceptable downtime) is met by the recovery pattern: backup and restore (hours), pilot light (tens of minutes), warm standby (minutes) or active-active (near zero). Infrastructure as code and automated runbooks shorten recovery.

Business continuity steps

1. Detect the failure through monitoring and alerts.
2. Declare an incident and follow the documented DR plan.
3. Fail over traffic using DNS or load-balancer health checks to the standby region.
4. Restore data from the latest clean version or snapshot.
5. Validate integrity, communicate with users, then fail back and run a post-incident review.
6. Test the plan regularly with DR drills.

Q4. Hybrid cloud for a government agency

Architecture

- Sensitive and regulated workloads stay on-premises in a private cloud or data center. Non-sensitive applications run in the public cloud.
- **Secure connectivity:** a dedicated private link (Direct Connect or ExpressRoute) or an IPsec site-to-site VPN. Network segmentation, firewalls, private endpoints and a zero-trust approach control east-west and north-south traffic.
- An API gateway and message broker mediate data exchange, and only the minimum required data crosses the boundary.

Identity management and encryption

- **Identity:** Federate the on-premises directory (Active Directory) with the cloud IdP using SAML or OIDC for single sign-on. Enforce MFA, role-based access control, least privilege and just-in-time privileged access.
- **Encryption:** TLS 1.2 or higher in transit. AES-256 at rest in both environments. Use a KMS/HSM with customer-managed keys, key rotation and separation of duties, ideally keeping the master keys on-premises (BYOK or HYOK).

Challenges

- Consistent security policy and compliance across two environments.
- Integration complexity, network latency and bandwidth cost.
- Unified monitoring, logging and visibility.
- Skill requirements, data governance and avoiding configuration drift.
- Cost management and keeping tools and versions compatible.

Q5. Multi-region high availability (SaaS provider)

Multi-region strategy

- Deploy the full stack in at least two regions, each spanning multiple availability zones. Use **active-active** for the lowest downtime, or active-passive (warm standby) to save cost.
- A global traffic manager (Route 53, Traffic Manager or Cloud DNS with health checks, or anycast) routes users to the closest healthy region.
- Keep the application tier stateless and defined as infrastructure as code, so regions are identical.

Data replication and failover

- Use multi-region databases (Aurora Global, Cosmos DB, Spanner) or asynchronous cross-region replication, with synchronous replication inside a region. Choose consistency levels to balance RPO against latency. Replicate object storage across regions.
- Configure health checks with sensible thresholds and short DNS TTLs. Failover can be automatic, with a promotion step for the replica database. Prevent split-brain with leader election or quorum, and test failover and failback regularly (chaos engineering).

Monitoring and alerting

- Collect metrics, logs and distributed traces centrally (CloudWatch, Prometheus/Grafana, ELK, Datadog).
- Track SLIs and SLOs (latency, error rate, availability, replication lag), and alert on thresholds and anomalies with on-call escalation (PagerDuty).
- Use synthetic probes from several locations, dashboards, and runbooks linked to alerts so problems are caught before users notice.

SET 2 (Big Data and Analytics)

Q1. Retail analytics in near real time

Architecture

- **Ingestion:** POS and e-commerce events go to a streaming service (Kafka, Kinesis or Pub/Sub). Batch sources arrive through ETL tools or CDC.
- **Storage:** A distributed data lake on object storage (S3, ADLS or GCS) in Parquet or Delta format.
- **Processing:** Spark for batch and Spark Structured Streaming or Flink for streams.
- **Serving:** A cloud data warehouse (Redshift, BigQuery or Snowflake), a low-latency store for recommendations, and BI tools.

Batch and stream integration

Use a Lambda or Kappa style design. The stream layer produces fresh, approximate results (such as live inventory), while the batch layer recomputes accurate history. A unified engine such as Spark, or table formats such as Delta or Iceberg, lets both write to the same tables.

Data lake and warehouse together

The lake keeps raw and semi-structured data cheaply (bronze, silver and gold layers). Curated, modeled gold data is loaded into the warehouse for fast SQL and reporting. A lakehouse design can merge the two.

Scalability and cost

- Separate storage from compute and autoscale clusters.
- Use spot instances, partitioning, compression and columnar formats.
- Apply lifecycle tiering and serverless query engines, and set budgets and alerts.

Q2. Secure healthcare cloud

Secure architecture

Use a dedicated virtual private cloud with private subnets, no public access to data stores, private endpoints, a WAF and DDoS protection. Medical images go to encrypted object storage (served through a PACS or VNA), and records go to an encrypted database. Choose compliant regions and sign a BAA or equivalent (HIPAA, GDPR, local law).

Encryption, IAM and key management

- **Encryption:** TLS in transit and AES-256 at rest for every store and backup.

- **IAM:** Single sign-on with MFA, role- and attribute-based access (doctors see only their patients), least privilege, and break-glass access with review.
- **Keys:** KMS or HSM with customer-managed keys, automatic rotation, separation of duties and key access logging.

Auditing, logging and compliance

Log every access to patient records immutably (CloudTrail or equivalent) in a write-once store and send logs to a SIEM. Run continuous compliance checks (Config, Security Hub), regular audits, and alerts on unusual access patterns.

Preventing unauthorized access and breaches

Zero-trust networking, network segmentation, data loss prevention, vulnerability scanning and patching, de-identification or masking for analytics, security training, incident response plans and regular penetration tests.

Q3. Serverless ticket booking

Design

A CDN serves the static front end. An API gateway receives requests and triggers functions (Lambda, Azure Functions or Cloud Functions) for search, booking and payment. A managed NoSQL or relational database (DynamoDB, Aurora Serverless) stores inventory and orders, and a cache protects it. A message queue (SQS or Pub/Sub) buffers booking requests, and workflows (Step Functions) coordinate payment and confirmation.

Event-driven scalability

Events (a request, a queue message, a database change) trigger functions that scale out automatically in parallel and back down to zero. Producers and consumers are decoupled, so a spike is absorbed by the queue instead of crashing the system.

Roles of components

- **Managed databases:** scale and replicate with little administration.
- **API gateway:** routing, authentication, throttling and caching.
- **Queues:** smooth bursts, retry failures and prevent overselling when combined with idempotent, conditional writes.

Advantages and limitations

- **Advantages:** automatic scaling, pay-per-use, no server management and faster releases.
- **Limitations:** cold starts, execution time and memory limits, vendor lock-in, harder debugging and local testing, possible database connection limits, and higher cost at constant heavy load.

Q4. ETL/ELT for a multinational company

Pipeline design

Extract from sales, finance, manufacturing and support systems using connectors, APIs, CDC and files (Fivetran, Glue, Data Factory). Land raw data in the lake. With **ELT**, load it into a cloud warehouse and transform it there with SQL tools such as dbt, producing staging, integrated and mart layers. Use ETL where data must be cleansed or masked before loading.

Quality, metadata and governance

- **Quality:** validation rules, deduplication, standardization, schema checks and automated tests (Great Expectations, dbt tests) with alerts.
- **Metadata:** a data catalog (Glue Catalog, Purview, Collibra) with lineage and business glossary.
- **Governance:** data owners and stewards, access control, masking of sensitive fields, retention policies and audit trails.

Workflow orchestration

Airflow, Azure Data Factory, Step Functions or Prefect schedule jobs, manage dependencies, retry failures, support backfills and send alerts, giving reliable and observable pipelines.

Performance and cost

Partitioning, clustering, columnar formats, materialized views, incremental loads, result caching, pruning unused data, storage tiering and compression, and separating workloads onto right-sized or auto-suspending compute.

Q5. Delivery delay prediction

Architecture

- **Ingest:** GPS streams through Kafka or Kinesis, weather through APIs, and shipment history through batch loads into a data lake.
- **Process:** Spark or Flink for stream and batch processing, with a feature store (SageMaker, Vertex AI or Feast) shared by training and inference.
- **ML:** managed training pipelines, a model registry, and real-time endpoints served through an API.

Big data for training and inference

Distributed frameworks (Spark MLlib, XGBoost on clusters, GPU instances) train on large historical data. Streaming jobs compute live features such as vehicle speed and route congestion, and models score events in real time or in batches.

Cloud-native AI/ML services

SageMaker, Azure ML or Vertex AI provide managed training, hyperparameter tuning, one-click deployment, autoscaling endpoints, A/B or canary releases, and model monitoring for drift and accuracy.

Ingestion, features and model updates

Use partitioned streaming ingestion with schema validation and a shared feature store to prevent training and serving skew. Retrain on a schedule or when drift is detected through MLOps CI/CD pipelines, with automated evaluation, versioning and rollback.

SET 3 (Migration, IoT, BI, Monitoring and Data Lifecycle)

Q1. Legacy migration for a manufacturing company

Strategy

Follow phases: assess and discover (inventory, dependencies, TCO), plan with a landing zone, migrate in waves (low-risk first), test and cut over, then optimize. Use migration tools (AWS MGN, DMS, Azure Migrate).

The six Rs applied

- **Rehost:** lift-and-shift VMs for quick moves of stable systems.
- **Replatform:** move the database to a managed service with few code changes.
- **Refactor:** rebuild critical apps as cloud-native microservices where the benefit is high.
- **Repurchase:** replace legacy CRM or HR with SaaS.
- **Retire:** decommission unused applications.
- **Retain:** keep apps that are not ready (compliance, latency to machines) on-premises for now.

Validating performance

Baseline metrics before migration, then compare after: load and stress tests, response time, throughput, database query times, user acceptance tests and monitoring dashboards.

Reducing risk and downtime

Continuous data replication with a short cutover window, pilot migrations, blue-green or phased cutover, a tested rollback plan, backups, security controls from day one and staff training.

Q2. Smart city IoT analytics

Architecture

Sensors connect through gateways using MQTT or CoAP to an IoT hub (AWS IoT Core, Azure IoT Hub). Messages flow into a broker or stream (Kafka, Kinesis), a stream processor (Flink, Spark Streaming) analyzes them, and results go to storage and dashboards.

Integration

The broker decouples devices from consumers. The stream processor filters, aggregates and detects anomalies, then writes hot data to a time-series database, raw data to a data lake, and alerts to notification services. Batch jobs analyze history.

High-volume, high-velocity data

Partitioned streams, edge computing to filter and aggregate near the source, compression, batching, windowing and autoscaling consumers.

Scalability, reliability and security

- **Scale and reliability:** managed, auto-scaling services, multi-AZ deployment, message retries and dead-letter queues.

- **Security:** per-device certificates, mutual TLS, secure provisioning, firmware updates, network segmentation and encryption at rest and in transit.

Q3. Executive BI dashboards

Architecture

Sources feed a pipeline into a lake and a cloud warehouse (Snowflake, BigQuery, Redshift), with a streaming path for real-time data. A semantic layer defines consistent metrics, and BI tools (Power BI, Tableau, Looker, QuickSight) present dashboards.

Integration with warehouses

Tools connect through native connectors or JDBC/ODBC, using either live queries or extracts and caching. The warehouse does the heavy computation, and the BI tool handles visualization.

Dashboard performance

Pre-aggregated tables and materialized views, incremental refresh, extracts or in-memory caching, partitioning and clustering, limiting visuals and filters, query pushdown and result caching.

Security

SSO and MFA, role-based access, row-level and column-level security by region, encryption, masking of sensitive fields, audit logs, and sharing controls with data loss prevention.

Q4. Gaming latency and performance

Monitoring solution

Collect metrics (CPU, memory, latency, player count), centralized logs and distributed traces with CloudWatch, Prometheus/Grafana, Datadog or OpenTelemetry. Add dashboards, SLOs and alerts.

Finding bottlenecks

- **Metrics** show which resource saturates and when.
- **Logs** explain errors and events.
- **Traces** follow a request across services to find the slow component, such as a database call or matchmaking service.

Resource optimization and capacity planning

Autoscaling with predictive and scheduled policies for daily peaks, right-sizing, load testing, trend analysis for forecasting, and caching to reduce database load.

Availability and cost

Multi-AZ and multi-region game servers placed close to players, load balancing with health checks, reserved or committed capacity for the baseline plus spot or on-demand for peaks, scaling down off-peak and budget alerts.

Q5. Financial data lifecycle management

Lifecycle strategy

Ingest, store, use, archive and delete, with each stage governed by policy. Automated lifecycle rules move data between storage tiers as it ages.

Classification, retention and archival

- **Classification:** tag data by sensitivity and regulatory category (PII, financial, public).
- **Retention:** keep records for the legally required period, then delete securely. Use legal hold when needed.
- **Archival:** hot storage for recent data, infrequent-access for months-old data, and archive tiers (Glacier, Archive storage) for older records, with object lock (WORM) for immutability.

Governance, auditing and compliance

Governance defines ownership and policies, and auditing provides evidence through immutable access logs. Compliance (SOX, PCI DSS, GDPR, Basel) requires lineage, retention proof and regular reviews, since failures lead to fines and loss of trust.

Balancing accessibility, security and cost

Tier by access frequency, compress and use columnar formats, query archives in place with serverless tools, encrypt all tiers with managed keys, apply least-privilege access, and review storage usage and costs regularly.